Lab 5: Identification and Removal of Outliers for Improving Data Quality

```
import pandas as pd
import numpy as np
from scipy import stats
import seaborn as sns
from sklearn.preprocessing import RobustScaler

from sklearn.datasets import load_breast_cancer

data=load_breast_cancer(as_frame=True).frame[['mean radius','mean
area']].dropna()
df=pd.DataFrame(data)
df
```

```
     mean radius  mean area
0          17.99     1001.0
1          20.57     1326.0
2          19.69     1203.0
3          11.42      386.1
4          20.29     1297.0
..           ...        ...
564        21.56     1479.0
565        20.13     1261.0
566        16.60      858.1
567        20.60     1265.0
568         7.76      181.0

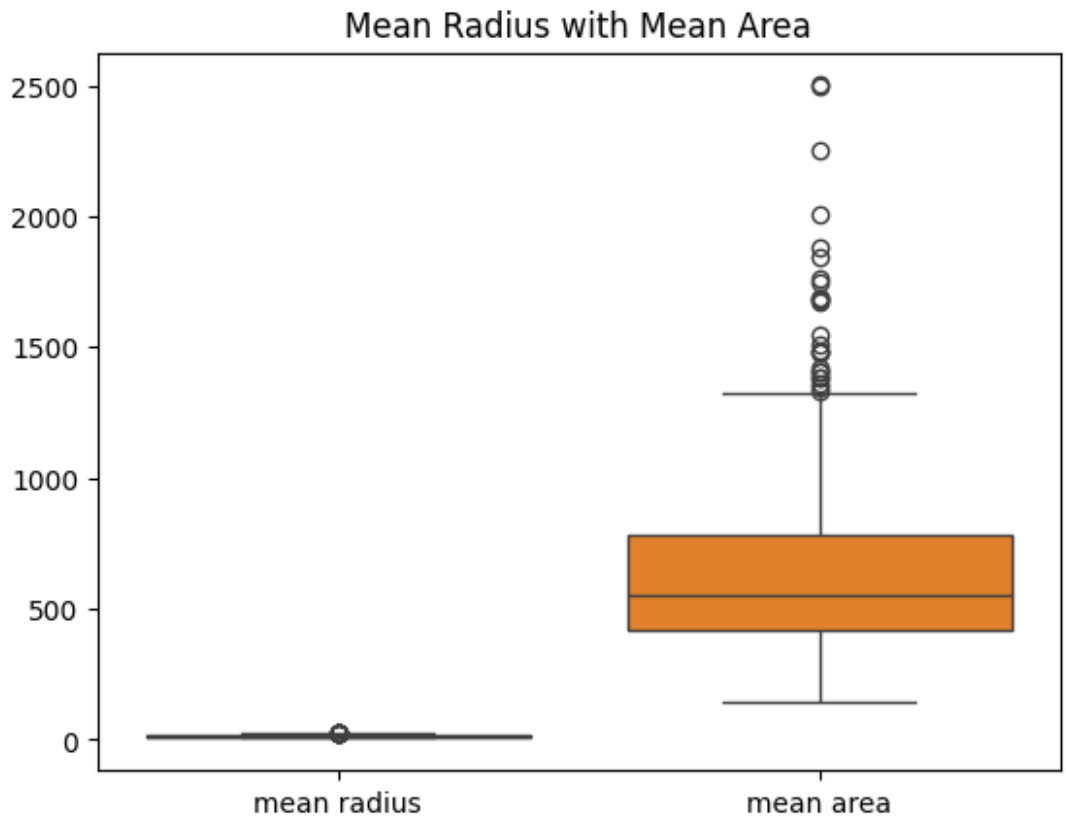
[569 rows x 2 columns]
```

Discover Outlier with boxplot

```
import seaborn as sns
import matplotlib.pyplot as plt

# sns.boxplot(x='mean radius', y='mean area', data=df)

sns.boxplot(df)
# plt.boxplot(x='mean radius', y='mean area', data=df)
plt.title("Mean Radius with Mean Area")
plt.show()
```



Identify Outliers Using Z-score

```
z_scores = np.abs(stats.zscore(df))
outliers_z= (z_scores >3)
outliers_z
```

```
array([[False, False],
       [False, False],
       [False, False],
       ...,
       [False, False],
       [False, False],
       [False, False]], shape=(569, 2))
```

IQR method

```
Q1 = df.quantile(0.25)
Q3 = df.quantile(0.75)
IQR = Q3 - Q1
outlier_iqr = ((df < (Q1 - 1.5 * IQR)) | (df > (Q3 + 1.5 * IQR)))
outlier_iqr
```

```
     mean radius  mean area
0          False      False
```

```
1          False      False
2          False      False
3          False      False
4          False      False
..           ...        ...
564        False       True
565        False      False
566        False      False
567        False      False
568        False      False

[569 rows x 2 columns]
```

Remove outliers

```
df_no_outliers = df[~((df < (Q1 - 1.5 * IQR)) |(df > (Q3 + 1.5 *
IQR))).any(axis=1)]
df_no_outliers
```

```
     mean radius  mean area
0          17.99     1001.0
1          20.57     1326.0
2          19.69     1203.0
3          11.42      386.1
4          20.29     1297.0
..           ...        ...
562        15.22      716.9
565        20.13     1261.0
566        16.60      858.1
567        20.60     1265.0
568         7.76      181.0

[544 rows x 2 columns]
```

Replace Outliters with Median

```
df_replace_median= df.copy()
for col in df.columns:
    median = df[col].median()
    df_replace_median[col] = np.where(outlier_iqr[col], median,
df[col])

df_replace_median
```

```
     mean radius  mean area
0          17.99     1001.0
1          20.57     1326.0
2          19.69     1203.0
3          11.42      386.1
```

```
4           20.29     1297.0
..            ...        ...
564         21.56      551.1
565         20.13     1261.0
566         16.60      858.1
567         20.60     1265.0
568          7.76      181.0

[569 rows x 2 columns]
```

Cap Outliers (winsorization)

```
df_winsorized = df.copy()
for col in df.columns:
    lower_limit = Q1[col] - 1.5 * IQR[col]
    upper_limit = Q3[col] + 1.5 * IQR[col]
    df_winsorized[col] = np.where(df[col] < lower_limit, lower_limit,
df[col])
    df_winsorized[col] = np.where(df_winsorized[col] > upper_limit,
upper_limit, df_winsorized[col])

df_winsorized
```

```
     mean radius  mean area
0          17.99     1001.0
1          20.57     1326.0
2          19.69     1203.0
3          11.42      386.1
4          20.29     1297.0
..           ...        ...
564        21.56     1326.3
565        20.13     1261.0
566        16.60      858.1
567        20.60     1265.0
568         7.76      181.0

[569 rows x 2 columns]
```

Log Transformation (only for positive values)

```
df_log=df.copy()
df_log['Log Mean Area'] = np.log(df_log['mean area'])
df_log
```

```
     mean radius  mean area  Log Mean Area
0          17.99     1001.0       6.908755
1          20.57     1326.0       7.189922
2          19.69     1203.0       7.092574
3          11.42      386.1       5.956096
```

```
4          20.29     1297.0       7.167809
..           ...        ...            ...
564        21.56     1479.0       7.299121
565        20.13     1261.0       7.139660
566        16.60      858.1       6.754721
567        20.60     1265.0       7.142827
568         7.76      181.0       5.198497

[569 rows x 3 columns]
```

Robust Scaling

s= (X - Median)/IQR

```
scaler= RobustScaler()
df_scaled= pd.DataFrame(scaler.fit_transform(df), columns=df.columns)
df_scaled
```

```
     mean radius  mean area
0       1.132353   1.241446
1       1.764706   2.138245
2       1.549020   1.798841
3      -0.477941  -0.455298
4       1.696078   2.058223
..           ...        ...
564     2.007353   2.560430
565     1.656863   1.958885
566     0.791667   0.847130
567     1.772059   1.969923
568    -1.375000  -1.021247

[569 rows x 2 columns]
```